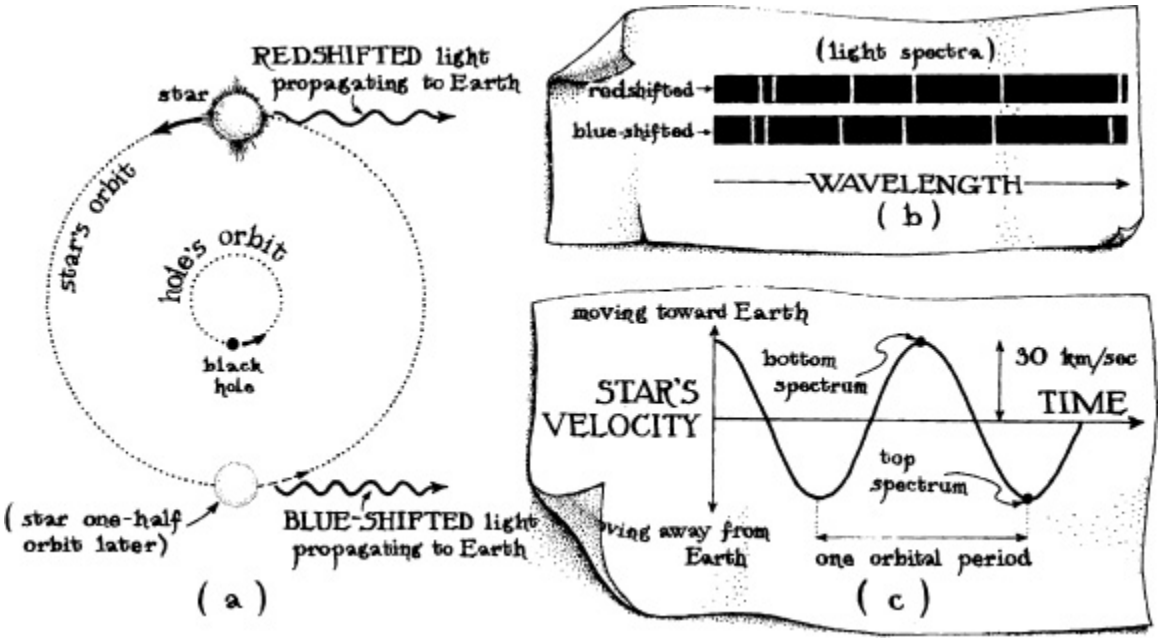
To search for a black hole, then, astronomers should look for a star whose spectra show a telltale periodic shift from red to blue to red to blue. Such a shift is an unequivocal sign that the star has a companion. The astronomers should measure the star's spectra to infer the velocity of the star around its companion, and from that velocity they should infer the companion's mass. If the companion is very massive and no light is seen from it at all, then the companion might well be a black hole. This was Zel'dovich's proposal.



8.3 Zel'dovich's proposed method of searching for a black hole. (a) The hole and a star are in orbit around each other. If the hole is heavier than the star, then its orbit is smaller than the star's as shown (that is, the hole moves only a little while the star moves a lot). If the hole were lighter than the star, then its orbit would be the larger one (that is, the star would move only a little while the hole moves a lot). When the star is moving away from Earth, as shown, its light is shifted toward the red (toward longer wavelength). (b) The light, upon entering a telescope on Earth, is sent through a spectrograph to form a spectrum. Here are shown two spectra, the top recorded when the star is moving away from Earth, the bottom a half orbit later when the star is moving toward Earth. The wavelengths of the sharp lines in the spectra are shifted relative to each other. (c) By measuring a sequence of such spectra, astronomers can determine how the velocity of the star toward and away from the Earth changes with time, and from that changing velocity, they can determine the mass of the object around which the star orbits. If the mass is larger than about 2 Suns and no light is seen from the object, then the object might be a black hole.

Although this method was vastly superior to any previous one, it nevertheless is fraught with many pitfalls, of which I shall discuss just two: First, the weighing of the dark companion is not straightforward. The star's